SPEAKORA

Complete Product & Technical Documentation

English speaking & learning platform — mobile app, FastAPI backend, and admin content studio.

Product name: Speakora

Document type: Product + Architecture + Feature + API Reference

Version: 1.0 (November 2026)

Audience: Founders, developers, QA, content editors

Repository: english development App (monorepo)

Table of Contents

1. Executive Summary
2. Product Vision & Brand
3. System Architecture
4. Technology Stack
5. Repository Structure
6. Learner Journey (End-to-End)
7. Mobile App Features
8. Learning Path & Lesson Unlock Rules
9. Exercises System
10. Speaking Practice (Voice Chat)
11. Role Plays (10 Scenarios)
12. Curriculum Content (A1 Unit 1)
13. Admin Panel (Content Studio)
14. Backend API Reference
15. Database & Progress Model
16. AI Pipeline (Groq)
17. Environment & Local Setup
18. Testing & Quality
19. Known Flows & Acceptance Criteria
20. Glossary

1. Executive Summary

Speakora is a mobile-first English learning product focused on structured CEFR lessons, interactive exercises, timed AI speaking practice, and scenario-based role-play voice chats. The platform has three surfaces:

- **Mobile app** (Expo / React Native) — learner experience
- **Backend API** (FastAPI / PostgreSQL) — auth, curriculum, progress, AI voice
- **Admin panel** (Next.js) — curriculum content studio

> **Current release focus:** A1 Unit 1 “Introducing Yourself” with 10 lessons, sequential unlock (exercises + speaking), 10 role-play scenarios (4 minutes each), branded splash, exercise feedback tones, and profile progress overview.

2. Product Vision & Brand

2.1 Brand

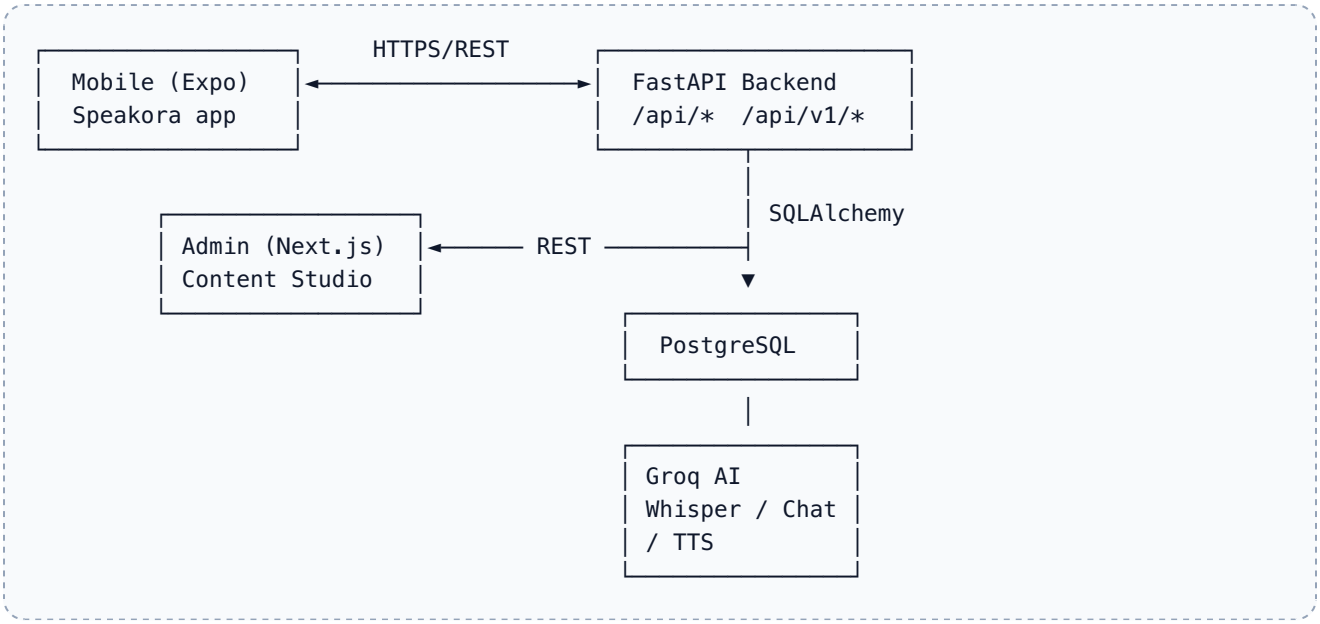
Item	Value
Product name	Speakora
Wordmark	Speak (white/blue) + ora (accent yellow)
Tagline	Speak English with confidence
Primary colors	Indigo / blue (#312E81, #2563EB), yellow accent (#FDE047)
Mascot	Friendly AI tutor robot (splash / onboarding)
App scheme	`speakora://`

Former working name “YayTalk / Yay” has been fully renamed to Speakora in product UI.

2.2 Learning principles

- Structured CEFR progression (starting at A1)
- Practice + immediate feedback (visual, text explanations, audio tones)
- Speaking first: every lesson ends with voice practice before unlocking the next
- Real-life role plays for transferable conversation skills
- Measurable progress: completed lessons, speaking time, streaks

3. System Architecture



3.1 High-level responsibilities

Layer	Responsibility
Mobile	Auth, onboarding, learning path, exercises, speaking UI, role plays, profile
Backend	Auth/JWT, curriculum serving, attempt scoring, progress unlock, AI STT/chat/TTS
Admin	Create/edit courses, units, lessons, exercises; publish workflow
Database	Users, curriculum, exercises, attempts, lesson progress, speaking sessions
AI (Groq)	Speech-to-text, tutor/role-play replies, optional TTS audio

4. Technology Stack

Area	Stack
Mobile	Expo ~58, Expo Router, TypeScript, Axios, Reanimated, expo-audio, Lottie, Lucide
Backend	Python, FastAPI, SQLAlchemy, Pydantic, Alembic, Pytest
Admin	Next.js App Router, Tailwind CSS, shadcn/ui
Database	PostgreSQL
AI	Groq (Whisper large-v3, chat model fallbacks, Orpheus TTS when enabled)

5. Repository Structure

```
english development App/
├── mobile-app/          # Expo learner app (Speakora)
├── backend/             # FastAPI API + Alembic + seeds/tests
├── admin-panel/         # Next.js Content Studio
├── docs/                # Product documentation (this PDF)
└── README.md            # Architecture blueprint
```

6. Learner Journey (End-to-End)

1. **Splash** — indigo branded screen with robot + Speakora wordmark + welcome audio
2. **Onboarding** — goals, level, focus, schedule, tutor voice, plan screens
3. **Auth** — register / login / Google / forgot & reset password
4. **Learning Path** — vertical unit trail; only current + completed lessons open
5. **Lesson exercises** — up to 3 tries; correct/incorrect tones; explanations
6. **Speaking** — 3-minute hold-to-talk AI voice practice
7. **Return to unit** — lesson marked complete → next lesson unlocks
8. **Role Plays** — optional 4-minute two-person scenario chats
9. **Profile** — lessons completed, speaking time, streak

7. Mobile App Features

7.1 Tabs

Tab	Purpose
Learning Path	Unit trail of lessons with lock/current/completed states
Role Plays	10 scenario cards → timed voice role play
Profile	Progress overview stats
Help	Support / help content

7.2 Splash & branding

- Native Expo splash uses indigo background + robot splash art
- In-app splash: gradient, robot portrait, pulse rings, floating motion, Speakora title
- Shared brand constants in `mobile-app/src/constants/brand.ts`

7.3 Auth screens

- Login, Register, Forgot password, Reset password

- Google Sign-In supported via configured web client id
- JWT stored in AsyncStorage (`access_token`)

8. Learning Path & Lesson Unlock Rules

> **Unlock rule (important):** A lesson unlocks the *next* lesson only after **all exercises are answered correctly** *and* the learner finishes that lesson's **speaking session**.

8.1 Path UI states

State	Meaning	UI
completed	`UserProgress.is_completed = true`	Green node with check
current	First incomplete lesson	Blue pulsing node (tappable)
locked	Later incomplete lessons	Pink/locked node (disabled)

8.2 Completion pipeline

1. Each correct unique exercise updates `LessonProgress` (exercise score).
2. When every exercise in the lesson is correct → `LessonProgress.is_completed = true`.
3. Speaking session `POST /api/v1/speaking/sessions/{id}/end` calls `mark_lesson_complete_after_speaking`.
4. If exercises are done, sets `UserProgress.is_completed = true` (path unlock).
5. Mobile speaking screen shows **Back to unit** and the Learning Path reloads on focus.

```
Greetings (exercises → speak) → unlock Introducing yourself
Introducing yourself (exercises → speak) → unlock Personal information
Personal information → … → Unit examination
```

9. Exercises System

9.1 Supported types

Type	Learner UI
multiple_choice / mcq	Tap option list
fill_blank	Text input
sentence_ordering	Tap tokens into order
matching / drag-drop-match	Match / drag chips

9.2 Attempt rules

- Maximum **3 tries** per exercise
- Wrong answers play negative tone + show remaining tries
- Correct answers play positive tone + show explanation
- After 3 fails, correct answer is revealed and learner can continue
- Attempts posted to `POST /api/v1/exercises/{id}/attempts`

9.3 Feedback tones

- Assets: `assets/audio/correct.*`, `assets/audio/incorrect.*`
- Helper: `mobile-app/src/lib/feedbackTones.ts` (preload + play + light vibration)

10. Speaking Practice (Voice Chat)

- Route: `/lesson/[id]/speak`
- Duration: **3 minutes** countdown timer
- Hold-to-talk microphone; audio uploaded as AAC/M4A
- Backend: create speaking session → message turns (STT → tutor reply → optional TTS) → end session
- On finish: return to Learning Path; unlock next lesson if exercises were completed

10.1 Speaking API

Method	Path	Purpose
POST	`/api/v1/speaking/sessions`	Start session (`lesson_id`)
POST	`/api/v1/speaking/sessions/{id}/messages`	Upload audio turn
POST	`/api/v1/speaking/sessions/{id}/end`	Close session; may set `lesson_completed`

11. Role Plays (10 Scenarios)

Tab → list of 10 illustrated scenarios → dedicated screen `/role-play/[id]`.

- **Format:** two-person voice chat (learner ↔ AI partner in character)
- **Duration:** 4:00 with centered timer above voice orb
- **Controls:** hold mic to talk; Exit chip / Exit chat / finished Exit
- **Orb:** voice-reactive waveform orb (mic metering + fallback animation)
- **AI:** `role_context` form field switches backend to role-play system prompt

11.1 Scenario catalog

#	Title	You	Partner
1	Ordering Coffee	Customer	Barista
2	At the Airport	Traveler	Check-in agent
3	Meeting a Friend	You	Friend
4	Hotel Check-in	Guest	Receptionist
5	Doctor Visit	Patient	Doctor
6	Job Interview	Candidate	Interviewer
7	Restaurant Dinner	Diner	Waiter
8	Clothes Shopping	Shopper	Shop assistant
9	Asking for Directions	Visitor	Local resident
10	Phone Support	Customer	Support agent

12. Curriculum Content (A1 Unit 1)

Course: A1 English · **Unit:** Introducing Yourself

#	Lesson	Focus	Seed exercises
1	Greetings	Matching, MCQ, speaking	3
2	Introducing yourself	Fill blanks, sentence building	3
3	Personal information	Questions and answers	3
4	Countries and nationalities	Vocabulary, matching	3
5	Jobs and professions	Picture matching, grammar	3
6	Asking questions	Ordering, fill, MCQ	3
7	Listening practice	Dictation-style / MCQ	3
8	Speaking practice	AI speaking prep	2+
9	Revision	Mixed exercises	3
10	Unit examination	Comprehensive test	seeded set

Example — Greetings (3 exercises)

1. Matching: Match the greeting → *Hello*
2. MCQ: What do you say in the morning? → *Good morning*

3. MCQ: Which phrase means goodbye? → *See you later*

Example — Introducing yourself (3 exercises)

1. Fill blank: My name _____ Sam. → *is*
2. Ordering: My / name / is / Lina
3. MCQ: Choose the correct sentence → *I am Tom.*

Seed script: `backend/scripts/seed_a1_unit1.py`

13. Admin Panel (Content Studio)

- Next.js App Router + Tailwind + shadcn/ui
- Branded as **Speakora Content Studio**
- Manages curriculum entities (courses, units, lessons, exercises, speaking scenarios)
- Content statuses: draft → generated → under_review → changes_requested → approved → published → archived
- Learner-visible statuses: `approved`, `published`

14. Backend API Reference

14.1 Auth

Method	Path	Notes
POST	`/api/auth/register`	Create account + token
POST	`/api/auth/login`	Email/password
POST	`/api/auth/google`	Google ID token
POST	`/api/auth/forgot-password`	Reset flow start
POST	`/api/auth/reset-password`	Reset with token
GET	`/api/auth/me`	Current user

14.2 Curriculum & content

Method	Path	Notes
GET	`/api/v1/courses`	Learning path with `is_completed`
GET	`/api/v1/lessons/{id}`	Lesson + exercises
GET	`/api/content/lessons/{id}/exercises`	Exercise list for player
POST	`/api/v1/exercises/{id}/attempts`	Score attempt + refresh progress

14.3 Progress

Method	Path	Notes
GET	`/api/v1/progress/overview`	lessons_completed, speaking seconds, streak
POST	`/api/v1/progress/sessions/end`	Generic timed session close

14.4 AI chat / role play

Method	Path	Notes
POST	`/api/ai/chat`	Audio (+ optional `lesson_id`, `role_context`)
POST	`/api/ai/speak`	TTS for a text line (opening lines)

14.5 Users / profile

Method	Path	Notes
GET/PATCH	`/api/v1/users/me/profile`	Onboarding profile fields

15. Database & Progress Model

15.1 Core tables (conceptual)

- `users` — accounts
- `courses` → `units` → `lessons` → `exercises`
- `exercise_attempts` — each submit
- `lesson_progress` — exercise score / exercises_completed flags
- `user_progress` — path completion + AI talk time (unlock source of truth)
- `speaking_sessions` / `speaking_messages` — voice turns
- `learning_sessions` — timed practice blocks for streaks/stats

15.2 Unlock source of truth

Learning Path reads `UserProgress.is_completed` via `GET /api/v1/courses`. Exercise-only completion updates `LessonProgress` but does *not* unlock the next node until speaking ends successfully with exercises done.

16. AI Pipeline (Groq)

1. **STT:** Whisper large-v3 transcription of uploaded audio
2. **Chat:** JSON reply with `reply` + `grammar_correction`

3. **Modes:** tutor correction prompt (lessons) vs role-play character prompt
4. **TTS:** Orpheus (optional; fails soft if unavailable)
5. **Audio format:** client records AAC/M4A; backend sniffs magic bytes before Whisper

> **Ops note:** Keep `GROQ_API_KEY` in `backend/.env` only. Never commit secrets. Accept TTS model terms in Groq console if voice playback is required.

17. Environment & Local Setup

17.1 Backend

```
cd backend
python -m venv venv && source venv/bin/activate
pip install -r requirements.txt
# configure backend/.env → DATABASE_URL, GROQ_API_KEY, JWT secret, etc.
alembic upgrade head
python scripts/seed_a1_unit1.py
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

17.2 Mobile

```
cd mobile-app
npm install
# .env → EXPO_PUBLIC_API_URL=http://<LAN-IP>:8000
npx expo start -c
```

On a physical phone, localhost is rewritten to the Expo LAN host automatically when `EXPO_PUBLIC_API_URL` uses localhost.

17.3 Admin

```
cd admin-panel
npm install
npm run dev
```

18. Testing & Quality

- Backend: Pytest suites for auth, content, exercises/progress, speaking, AI chat (mocked Groq)
- Mobile: TypeScript (`tsc --noEmit`)
- Workspace rules: TypeScript everywhere on clients; async FastAPI; Axios on mobile; Tailwind + shadcn on admin

19. Known Flows & Acceptance Criteria

19.1 Lesson unlock acceptance

1. Open Learning Path → only first incomplete lesson is tappable
2. Complete all Greetings exercises (tones play) → Speaking opens
3. Finish speaking → **Back to unit**
4. Greetings shows completed (green); Introducing yourself becomes current (blue)
5. Repeat for Introducing yourself → unlocks Personal information

19.2 Role play acceptance

1. Open Role Plays → see 10 image cards
2. Start a scenario → timer centered above orb
3. Hold mic → partner replies in character
4. Exit available during chat and after finish

19.3 Branding acceptance

1. Native/loading splash is indigo with robot (not plain white name-only)
2. In-app splash shows Speakora robot + animated presence
3. Admin titles show SPEAKORA / Speakora Content Studio

20. Glossary

Term	Meaning
CEFR	Common European Framework of Reference (A1–C2)
LessonProgress	Exercise scoring state for a lesson
UserProgress	Path unlock / completion flag + talk time
Role Play	Timed two-person AI scenario conversation
Speaking session	Lesson voice practice block that can complete a lesson
STT / TTS	Speech-to-text / text-to-speech

Document generated for the Speakora monorepo (mobile-app · backend · admin-panel). For deeper architecture history and multi-release roadmap, also see repository `README.md`.